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GOPEL

GOVERNANCE ORCHESTRATOR POLICY ENFORCEMENT LAYER

CANONICAL DESCRIPTION · VERSION 1.4 · MARCH 2026

Full Name	Governance Orchestrator Policy Enforcement Layer
Classification	Published Reference Implementation — Non-Cognitive Multi-AI Governance Layer
Also Termed	Deterministic Multi-AI Governance Control
Version	v1.4 · Produced under two rounds of seven-platform CAIPR adversarial review
Date	March 2026
CBG Authority	Checkpoint-Based Governance v4.7
Framework	HAIA-RECCLIN 2026 Edition
Repository	github.com/basilpuglisi/HAIA · CC BY-NC 4.0
Book	Governing AI: When Capability Exceeds Control · ISBN 9798349677687
Attribution	#AIassisted under HAIA-RECCLIN & Checkpoint Based Governance

GOPEL is the only published, fully disclosed reference implementation of a non-cognitive multi-AI governance architecture anywhere in the world. In 2025, while building the HAIA-RECCLIN governance framework, the search for a pre-inference enforcement layer came up empty. In January 2026, during the HAIA-Agent architecture work, the question was put directly to multiple AI platforms under CAIPR parallel dispatch: does anything like this exist? The answer across every platform was the same. Nothing had been published. GOPEL was built to fill that gap.

A prior art search conducted across AI governance literature, confidential computing standards, and multi-agent orchestration frameworks as of March 2026 found no identical architecture in any published specification, open-source repository, academic paper, or policy framework. GOPEL was not built to be first. It was built because the gap was a governance failure waiting to happen.

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GOPEL — Full Architecture at a Glance

v1.4 · March 2026 · #AIgenerated under HAIA-RECCLIN governance

Tier 0 – Human Arbiter | GOPEL – Non-Cognitive | External – Navigator | External – Gov. Monitoring | Policy / CPE

HUMAN ARBITER
TIER 0

Sets operating model, dispatch mode, data sensitivity, platform selection policy.
All governance artifacts (trust anchors, routing matrices, thresholds, allowlists) established here and loaded into GOPEL as configuration inputs.

GOPEL BOUNDARY INTEGRITY CHECK
PRE-OPERATION

Container verified. Source modules unaltered. Non-cognitive constraint confirmed.
Liveness detection circuit active. Fail closed if check fails.

DISPATCH CONFIGURATION
CONFIGURATION

RECCLIN-formatted prompt assembled by human or dispatch config (not by GOPEL).
CPE profile assigned. Data sensitivity classified. Endpoint capability verified.
Profile 0 = system default (blocks unverified or unclassified dispatches).

TARGETED ROLE DISPATCH
RECCLIN STRATEGY

Pre-approved routing matrix executed.
No runtime capability evaluation.
Navigator: role synthesis only.
No convergence analysis.

CAIPR PARALLEL DISPATCH
PARALLEL

Identical prompt to 3/5/7/9/11 platforms.
Independence enforced throughout.
Navigator: convergence analysis.
Cross-platform bias detection.

GOPEL — Seven Deterministic Operations

1. **DISPATCH** — Authenticated, signed dispatch envelope, CPE-classified
2. **COLLECT** — Complete, unedited, signed envelope verified on return
3. **ROUTE** — All responses out to Navigator (EXTERNAL)
4. **LOG** — Six record types, append-only, hash-chained, signed
5. **PAUSE** — Preconfigured gate: Model 1 (end) / Model 2 (each role)
6. **HASH** — Post-quantum signature agility applied to all records
7. **REPORT** — Approval rate, reversal rate, consecutive agreement run, and zero-reversal count — deterministic counters only

Route → external

NAVIGATOR
EXTERNAL – OUTSIDE GOPEL BOUNDARY

Lives in: AI platform / NotebookLM / custom in-house model / human.
CAIPR path: Convergence analysis. Conflict identification. Dissent preserved.
Targeted Role path: Role synthesis only. No convergence analysis.
All recommendations labeled AI-generated.
All analytical content originates here, never in GOPEL.

Log → unchanged

GOPEL LOG — Navigation Record Written

Navigator synthesis logged exactly as received. No modification by GOPEL.

HUMAN ARBITER — CBG CHECKPOINT
CBG CHECKPOINT

Receives: raw outputs + Navigator synthesis + dissent + decision point elevated for review.
Decides: Approve / Modify / Reject.
Arbitration Record + Decision Record written. Hash chain advanced.

GOPEL REPORT OPERATION — Four Deterministic Counters
REPORT OPERATION

Approval rate / Reversal rate / Consecutive agreement run / Zero-reversal count.
Deterministic arithmetic operations on Decision Records only.
GOPEL counts, it does not infer.

GOVERNANCE MONITORING LAYER
EXTERNAL – OUTSIDE GOPEL BOUNDARY

Interprets counts against configured thresholds.
Triggers audits and escalations when thresholds are crossed.
Delivers alerts to human authority responsible for acting on them.

AUTHORIZED OUTPUT — COMPLETE AUDIT CHAIN
AUTHORIZED OUTPUT

Reconstructable from Request Record to Decision Record.
Cryptographic integrity verifiable by any external auditor.

CPE — Four Data Protection Profiles

PROFILE 0 — SYSTEM DEFAULT — PROHIBITION
Blocks any dispatch lacking a valid endpoint capability flag or recognized data sensitivity label.
No regulated data moves without explicit authorization.

PROFILE 1 — ATTESTED CONFIDENTIAL — LEGAL STANDARD
TEE attestation verified against approved trust anchor.
Legal standard required by HIPAA for PHI.
VAISA is the legislative mandate mechanism.

PROFILE 2 — MINIMIZED EXTERNAL — INTERIM DEFAULT
Deterministic tokenization of structured identifiers.
Operational default until Profile 1 is available.
Does not cover free text or unstructured content.

PROFILE 3 — HUMAN-GATED EMERGENCY — EXCEPTION ONLY
Named arbiter explicit approval required.
72-hour limit. Post-use review within 14 days.
Emergency exception path only.

Key Architectural Boundaries

- The Navigator is always external to GOPEL – GOPEL routes to it and logs its output unchanged.
- The governance monitoring layer is always external to GOPEL — GOPEL counts; the monitoring layer interprets.
- All analytical content and escalation decisions originate outside GOPEL.
- **GOPEL moves, logs, and enforces.**

#AIgenerated under HAIA-RECCLIN governance — Human arbiter review and approval required before any external distribution or citation.

CANONICAL DESCRIPTOR

GOPEL is a Published Reference Implementation of a Non-Cognitive Multi-AI Governance Layer, also termed Deterministic Multi-AI Governance Control. GOPEL is a pre-inference enforcement layer for governed multi-AI workflows.

All governance artifacts that GOPEL executes — including trust anchors, endpoint capability flags, data sensitivity labels, measurement allowlists, routing matrices, threshold configurations, and platform anchor assignments — are established by human or enterprise governance authority outside GOPEL and loaded into GOPEL as configuration inputs. GOPEL executes configured policy. GOPEL does not generate, evaluate, or modify the policies it enforces.

POINT 1 GOPEL Is Governed Infrastructure

GOPEL is infrastructure. Specifically, it is the secured, governed channel between the human arbiter and every AI platform the human uses. Every request to an AI platform passes through GOPEL, and every response returns through GOPEL. Nothing leaves or enters the governed boundary without cryptographic integrity verification, authenticated identity binding, and a logged record.

The security and the governance are distinct properties that GOPEL provides simultaneously. The security property means no data moves without cryptographic verification and authenticated identity binding. The governance property means no operation executes without a logged record, no checkpoint advances without human approval, and no authorization proceeds outside the enterprise policy framework.

> Security without governance is a locked room with no accountability for what happens inside; governance without security is a chain of custody with gaps. GOPEL provides both. The canonical descriptor names the governance function as primary because governance is the purpose, and security is the mechanism that makes the governance trustworthy.

GOPEL governs but does not think. Policy application and enforcement are deterministic for any given configuration, evidence set, and transaction state. GOPEL materially reduces cognitive attack surface because it exposes no content reasoning or judgment path. The attack surface reduces to message transport, cryptographic operations, configuration integrity, key management, and attestation policy — all of which are deterministic and auditable. This is not a limitation of the architecture. It is the security design.

POINT 2 GOPEL Carries RECCLIN-Formatted Requests and Returns Structured Outputs

GOPEL carries prompts. GOPEL does not write prompts, evaluate prompts, or modify prompts. This distinction is non-negotiable because formatting a prompt requires content evaluation, and content evaluation violates the non-cognitive constraint that makes GOPEL trustworthy.

What GOPEL carries are prompts that the dispatch configuration has already formatted with RECCLIN governance instructions before GOPEL receives them. Those instructions require every AI platform response to include the following structure: role, task, answer, sources, conflicts, confidence, expiry, fact-to-tactic-to-KPI, recommendation, and decision point.

Every governed transaction using this response schema results in all six audit record types through the Seven Deterministic Operations. The RECCLIN format is the governance schema that makes outputs comparable, auditable, and reviewable across platforms. GOPEL validates that the schema is present in every dispatch and logs every response in full, unedited, exactly as received. GOPEL validates structure; GOPEL does not evaluate content.

POINT 3 GOPEL Supports Three Agent Operating Models

The agent is distinct from GOPEL. GOPEL is the infrastructure. The agent is the mode of operation that determines how GOPEL's outputs flow and where CBG checkpoints interrupt the workflow. Choosing the operating model is a CBG decision made by the human arbiter before any workflow begins, and the model selection is logged as part of the Request Record.

The Navigator is a cognitive function that lives outside GOPEL entirely. The Navigator is an AI platform, an internal tool such as NotebookLM, or a custom in-house model that receives the collected platform responses from GOPEL's Route operation and produces convergence analysis and synthesis. GOPEL routes to the Navigator, and GOPEL logs what the Navigator returns as a Navigation Record. GOPEL never performs synthesis, generates recommendations, or evaluates content — the Navigator does the cognitive work while GOPEL moves, logs, and enforces.

MODEL 1 Agent Responsible AI

One CBG checkpoint at workflow end

GOPEL dispatches to all platforms, collects all responses, routes to Navigator (external).

Navigator produces synthesis. GOPEL logs as Navigation Record.

Governance package assembled: raw outputs + synthesis + decision point.

Designed for routine, lower-risk operations where governance at operational speed is the priority.

MODEL 2 Agent AI Governance

CBG checkpoint at every RECCLIN role gate

GOPEL dispatches, collects, routes to Navigator at every role gate.

Human receives governance package at each gate; approves before workflow advances.

Designed for high-risk decisions: healthcare, credit, employment, law enforcement.

Operationalizes EU AI Act Article 14 — oversight proportional to risk.

MODEL 3 Manual Human AI Governance

No agent automation

Human orchestrates every interaction manually across multiple AI platforms.

GOPEL logs operations but does not automate dispatch, routing, or synthesis.

Human performs the Navigator function directly.

Model that produced the published book and all operational case studies.

POINT 4 GOPEL Supports Targeted Role Dispatch (RECCLIN Strategy)

When a task has distinct functional components that map to RECCLIN roles, GOPEL can dispatch different parts of the task to different platforms. Targeted Role Dispatch applies a pre-approved routing matrix established outside GOPEL by governance policy; GOPEL executes the configured route and does not evaluate comparative platform capability at runtime.

> Targeted Role Dispatch does not produce convergence analysis. Platforms answer different questions rather than the same question; there is no independent replication of the same task, and therefore no cross-platform bias detection applies. The audit record for a Targeted Role Dispatch session must explicitly identify the mode used. Targeted Role Dispatch produces role-specialized excellence; CAIPR Parallel Dispatch produces cross-validated evidence. These are complementary capabilities, not interchangeable ones.

POINT 5 GOPEL Supports CAIPR Parallel Dispatch

When the task requires cross-validated evidence rather than role specialization, GOPEL dispatches the identical prompt to multiple independent AI platforms simultaneously. Platform count follows the odd-number protocol: 3, 5, 7, 9, or 11 platforms. Odd numbers prevent tie outcomes when convergence thresholds are applied. The minimum for any CAIPR session is three platforms.

Parallel dispatch preserves independence. No platform receives any other platform's output before producing its own. Sequential dispatch is prohibited under CAIPR because it contaminates independence: Platform B, having seen Platform A's output, no longer generates independent evidence, and convergence analysis is only valid when all platforms answered the same question without visibility into each other's responses.

> Consensus thresholds decline as platform count increases to preserve dissent: 2 of 3 at three platforms, 3 of 5 at five, 4 of 7 at seven, 5 of 9 at nine, and 6 of 11 at eleven. Dissent is governance data. A single platform disagreeing with all others is not a noise event to suppress; it is the signal that prevented documented governance failures in operational practice.

POINT 6 GOPEL Cuts Off Communication When Security Is Compromised

GOPEL operates under a fail-closed security architecture. When any security verification fails, GOPEL does not proceed and does not silently degrade. The pipeline halts, the failure event is logged, and the human arbiter is alerted.

Three layers of security verification operate before dispatch and upon response collection for every transaction: container integrity check (GOPEL verifies its own boundary before any operation executes), API security check (endpoint must meet minimum security requirements for the data sensitivity classification of the dispatch), and transport integrity check (signed dispatch envelope verified on every response collected).

A circuit breaker halts the entire pipeline when configurable thresholds are exceeded. Once a CRITICAL breach is logged, the pipeline cannot resume until a specifically authorized human arbiter executes the acknowledge-breach procedure, preserving the elevated governance posture even after the pipeline resumes.

POINT 7 GOPEL Produces a Cryptographic Audit Trail, Not Just a Log

The term 'log' understates what GOPEL produces. Every operation writes to an append-only, hash-chained, digitally signed audit trail. Append-only means nothing is overwritten. Hash-chained means each record contains the SHA-256 hash of the previous record — any alteration to any record breaks the chain. This is cryptographic chain of custody that produces forensically useful evidence suitable for regulatory audit and legal proceedings.

Six record types are produced for every transaction: Request Record, Dispatch Record, Response Record, Navigation Record, Arbitration Record, and Decision Record. The Arbitration Record proves that a human decided — not merely that a human was present. The Decision Record links the final authorized output to every upstream record in the chain.

> The post-quantum cryptographic agility amendment extends the signature architecture to remain valid across multi-decade retention obligations. ECDSA P-256 and Ed25519 are vulnerable to Shor's algorithm on a cryptographically relevant quantum computer. The amendment replaces these with post-quantum signature standards consistent with NIST post-quantum cryptography guidance and does not alter the seven operations or the non-cognitive constraint.

POINT 8 CPE Governs What Happens to Data Inside the AI Platform

After GOPEL dispatches a prompt and before GOPEL collects the response, the data is inside an AI platform's inference stack. GOPEL has zero visibility into that interval. The Confidential Processing Extension classifies every dispatch into exactly one of four profiles before any data moves, based on two deterministic inputs: the endpoint capability flag and the data sensitivity label.

PROFILE 0 PROHIBITION

SYSTEM DEFAULT

System default. Blocks any dispatch lacking a valid endpoint capability flag or recognized data sensitivity label.

No regulated data moves to an unverified endpoint without explicit human arbiter authorization documented in the audit trail.

PROFILE 1 LEGAL STANDARD

ATTESTED CONFIDENTIAL

TEE attestation verified against approved trust anchor.

Legal compliance standard required by HIPAA for PHI and equivalent regulated material.

VAISA is the legislative mandate mechanism at the inference layer.

PROFILE 2 INTERIM DEFAULT

MINIMIZED EXTERNAL

Deterministic tokenization of structured regulated identifiers before dispatch.

Token map never leaves GOPEL custody. Operational default for tokenized regulated structured data.

Does not solve free text leakage or unstructured document sensitivity.

PROFILE 3 EXCEPTION ONLY

HUMAN-GATED EMERGENCY

Named arbiter explicit approval with documented justification required.

72-hour limit. Mandatory post-use review within 14 days.

Emergency exception path only. Not a compliance substitute for Profile 1.

> Operating under Profile 2 with raw PHI or classified material is a compliance gap, not a compliant posture. Profile 1 is the legal standard that HIPAA and equivalent frameworks already require at the inference layer. VAISA is the legislative mechanism designed to mandate what those frameworks already require. As of March 2026, most commercial LLM APIs do not yet expose attestation evidence at Profile 1 level.

POINT 9 Automation Bias Counting Is Built Into the Report Operation

The Report operation counts governance events. A separate governance monitoring layer — which resides outside GOPEL as a policy and oversight component — interprets what those counts mean and delivers alerts to the human authority responsible for acting on them. GOPEL counts; it does not infer.

GOPEL's Report operation produces four counters for every observation window: approval rate, reversal rate, consecutive agreement run, and zero-reversal count. These are deterministic arithmetic operations on Decision Records.

`Approval rate > 95%`	Governance monitoring layer triggers mandatory sampling audit
`Reversal rate < 2%`	Governance monitoring layer triggers mandatory sampling audit
`Consecutive agreement run > 20`	Governance monitoring layer triggers escalation regardless of aggregate rates
`Zero-reversal count >= 10`	Governance monitoring layer triggers automatic escalation regardless of all other metrics

> The automation bias monitoring exists because the most common and least visible governance failure is not the system doing something wrong. It is the human gradually stopping oversight without any single moment of failure that triggers an alert. The Report operation surfaces that pattern structurally.

POINT 10 Platform Selection Uses Anchor-Plus-Rotation to Prevent Systematic Bias

GOPEL executes platform selection for each session through an anchor-plus-rotation protocol. Anchor assignments and performance assessments are established by external governance policy and loaded into GOPEL as configuration inputs; GOPEL applies the preconfigured assignment and does not evaluate platform capability at runtime.

No platform holds a permanent anchor position across all sessions. The rotation requirement is a governance mechanism, not an efficiency preference. If Platform A always anchors the Research role and has systematic blind spots in a particular domain, those blind spots become structural in every research output that passes through GOPEL. Rotation distributes the anchor responsibility so that no single platform's limitations become the framework's limitations.

The cryptographic rotation seed means no external actor can predict which platforms GOPEL will select for any given dispatch. Predictable selection enables targeted adversarial manipulation of anchor platforms; cryptographic rotation materially reduces that attack vector without any cognitive evaluation.

POINT 11 GOPEL Returns Governance Packages Structured to the Operating Model and CBG Checkpoint

GOPEL's return path is as governed as its dispatch path. What returns to the human is not a raw AI response. It is a structured governance package that GOPEL assembles by collecting raw platform outputs, logging the Navigator synthesis exactly as received, and wrapping both into a structured package for the specific CBG checkpoint at which the human will review it.

All analytical fields in the governance package — synthesis, convergence map, confidence scores, recommendation, and the decision point elevated for checkpoint review — originate in the external Navigator function or the human arbiter, never in GOPEL. GOPEL wraps and routes; the Navigator produces the synthesis; the human exercises authority.

In all three models, the CBG checkpoint is the destination. The governance package exists to give the human everything needed to exercise genuine authority: not just presence in the workflow, but structural verification that the human's decision is the decision that governs the output. The Arbitration Record is the evidence that a human decided; the Decision Record is the evidence of what they decided.

KEY ARCHITECTURAL BOUNDARIES

1. The Navigator is always external to GOPEL. GOPEL routes to it and logs its output unchanged.
2. The governance monitoring layer is always external to GOPEL. GOPEL counts; the monitoring layer interprets.
3. All analytical content and escalation decisions originate outside GOPEL.
4. **GOPEL moves, logs, and enforces.**

PUBLISHED RESOURCES

GitHub	github.com/basilpuglisi/HAIA · CC BY-NC 4.0
Book	Governing AI: When Capability Exceeds Control · Basil C. Puglisi, MPA · ISBN 9798349677687
Congressional Pkg	AI Provider Plurality v9 · Senate Commerce / House Energy and Commerce / Senate Judiciary
Academic Paper	HAIA-RECCLIN Academic Working Paper — EU Regulatory Compliance Edition
VAISA	Verified AI Inference Standards Act — Legislative mandate mechanism for inference-layer attestation
Case Studies	006 — HAIA-CAIPR Protocol · 002 — CBG Audit Log · 001 — Thought Leader
Website	basilpuglisi.com

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